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Reduced Time to Surgery Improves Patient-Reported Outcome After Achilles Tendon Rupture

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Background: Patient outcome after an acute Achilles tendon rupture (ATR) continues to be suboptimal and heterogeneous. Thus, prognostic factors are called for to optimize evidence-based ATR treatment protocols; however, the influence of delayed time from injury to surgery (TTS) on patient outcome after ATR remains largely unknown.

Purpose: To determine whether patient outcomes and adverse events after surgical repair of acute ATR are related to delayed TTS.

Study Design: Cohort study; Level of evidence, 3.

Methods: Two hundred twenty-eight patients with ATR who were treated with uniform anesthetic and surgical techniques, within 10 days after injury, were retrospectively assessed. TTS depended on availability of the operating theater, and neither surgeon nor patient could influence the TTS. Patients were assigned to 1 of 3 groups: short TTS (<48 hours), intermediate TTS (48-72 hours), and long TTS (>72 hours). Patient-reported outcome at 1 year was assessed with the validated Achilles tendon Total Rupture Score, with scores higher than 80 on a 0- to 100-point scale indicating an overall good outcome. The incidences of adverse events (perioperative and postoperative) and deep venous thrombosis were assessed.

Results: Short TTS was significantly associated with increased rate of good outcome and reduced risk of adverse events. Seventy-one percent (95% CI, 60%-83%) of the patients with short TTS attained a good outcome compared with 44% (95% CI, 33%-56%) of the patients with long TTS (P = .002), with the intermediate TTS group in between, 63% (95% CI, 47%-78%). The incidence of adverse events was significantly reduced among patients with short TTS, 1.4% (95% CI, 1%-4%) compared with those with intermediate TTS, 11% (95% CI, 2%-21%) (P = .035) and those with long TTS, 14.8% (95% CI, 7%-23%) (P = .003). The risk of sustaining a deep venous thrombosis was not statistically significantly different among the 3 groups (P = .15).

Conclusion: Patients with acute ATR undergoing operation within 48 hours after injury had better outcomes and a lower number of adverse events compared with patients undergoing operation after 72 hours. These results align with evidence-based recommendations from other surgical disciplines and should be used as guidelines for optimizing ATR treatment protocols.

Registration: NTC01317160 (ClinicalTrials.gov).

Keywords: foot; Achilles tendon; rupture; time to treatment; patient-reported outcome measures; postoperative complications; venous thrombosis

The increasing incidence of Achilles tendon rupture (ATR)[8,9,19] further stresses the demands of evidence-based treatment protocols to optimize ATR patient care. Discussions for optimizing ATR patient care have mostly focused on whether to treat the patients surgically or nonsurgically, without considering a plausible negative effect of delayed time to surgery (TTS) on patient outcome.

An adverse effect of delayed TTS on patient outcome has been a common denominator in several orthopaedic fields; studies have examined long-term TTS (weeks to months) in rotator cuff tears[7] and meniscal and chondral injuries[3,13] and short-term TTS (days) in hip fractures.[12,18,21,25] In hip fracture surgery today, it is recommended not to delay surgery more than 48 hours after admission, because increases in mortality and both minor complications (eg, postoperative

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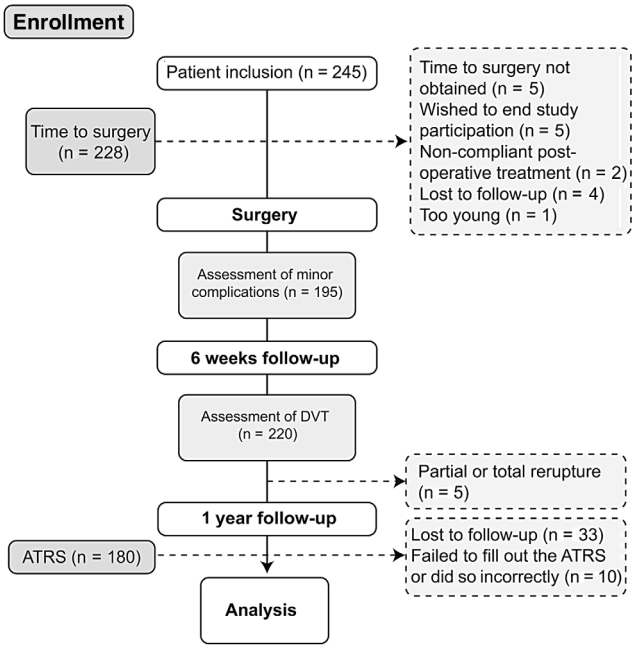

Figure 1. Patient flowchart: 228 patients with unilateral Achilles tendon rupture were included and then underwent surgery. ATRS, Achilles tendon Total Rupture Score; DVT, deep venous thrombosis.

infections) and major complications (eg, venous thromboembolism) are seen with prolonged delay of surgery.[14,18,21,25]

Subjective and functional outcomes after ATR vary greatly between patients; the risk of deep venous thrombosis (DVT) is high, and factors that influence the variability in outcomes are not fully understood.[4,5] As well, the relationships between delayed TTS, complications, and patient outcomes after ATR are mostly unknown.[17]

The purpose of this retrospective cohort study was to assess TTS in relation to complications and to validated patient-reported outcome measures 1 year after surgery. We hypothesized that delayed TTS may negatively affect patient outcome after surgical repair of an ATR and increase the rate of complications such as DVT and adverse events such as infections, surgical complications, and reoperations.

METHODS

Ethical approval was obtained from the Regional Ethical Review Committees in Sweden (Dnr: 2013/1791-31/3, 2009/2079-31/2), and the study was listed in clinicaltrials.gov.

Patients

Two-hundred forty-five patients who had undergone surgery after acute ATR injury were retrospectively included from 2 prospectively designed randomized controlled trials (RCTs) by Domeij-Arverud et al[6] and Valkering et al.[23] The RCTs investigated different postoperative rehabilitation protocols during the first 2 postoperative weeks compared with plaster cast, further described below in the section on postoperative treatment. These RCTs used identical protocols regarding inclusion criteria, exclusion criteria, anesthesia, and surgical technique. Both RCTs solely examined ATR patients who had undergone surgery after their injury, and there were no control groups with patients receiving nonoperative treatment. Written consent was collected from all patients in both RCTs at study inclusion. Patient inclusion and follow-up are described in Figure 1, and patient characteristics are described in Table 1.

Patients who had sustained an acute unilateral rupture at the midsubstance level of the Achilles tendon and had received surgery within 10 days from injury were eligible for inclusion. Exclusion criteria were current anticoagulation treatment (including high-dose acetylsalicylic acid), known kidney failure, heart failure with pitting edema, thrombophlebitis, thromboembolic event during the previous 3 months, known malignancy, hemophilia, pregnancy, other surgery during the previous month, inability to follow instructions, or planned follow-up at another hospital.

Time to Surgery

Patients were immediately assigned for surgery but had to wait for a free slot in the operation theater. Neither the patient nor the surgeon was able to affect the time from ATR injury to surgery. TTS (ie, the time from ATR injury to the start of the surgical procedure) was calculated by using the time point at which the patient sustained the injury, as described in the patient record, as well as the starting time point of the surgery as registered in the computerized operation report. Two-hundred twenty-eight patients with valid TTS received surgery at a major hospital in Sweden. The TTS variable was trichotomized according to the 48-hour time point chosen from other studies[20,21,25] plus an additional practical 24-hour cycle, resulting in the 72-hour time point, which is an important biological phase in tendon healing.[2] Patients were assigned to 1 of 3 groups; TTS less than 48 hours (short), TTS 48-72 hours (intermediate), and TTS more than 72 hours (long). After ATR surgery, patients were randomized to 3 different postoperative treatments, further described below.[6,23] Minor surgical complications, reruptures, and postoperative infections were summarized as adverse events. At 2 and 6 weeks postoperatively, the patients were examined for postoperative infections and additionally screened for DVT in the operated leg. At the 12-month follow-up, patient-reported outcome was assessed by use of the validated Achilles tendon Total Rupture Score (ATRS) questionnaire.[15]

Time to Surgery and Patient Characteristics

Among the patient characteristics (sex, age, and body mass index), only sex was statistically significantly associated with TTS (R = 0.15, P = .023); however, sex was not

TABLE 1
Patient Characteristics and Outcome[a]

Variable	Short TTS, <48 h (n = 74)	Intermediate TTS, 48-72 h (n = 49)	Long TTS, >72 h (n = 105)	*P* Value
Patient characteristics				
TTS, h:min, mean (SD)	34:58 (10:28)	64:47 (6:26)	105:15 (34:07)	NA
Sex, male, n (%)	60/74 (81)	47/49 (96)	77/105 (73)	**.004**
Age, y, mean (SD)	39 (7.2)	41 (8.7)	41 (8.5)	.37
BMI, kg/cm^2, mean (SD)	26.4 (3.4)	26.6 (2.7)	25.8 (3.5)	.26
Outcome variables				
ATRS >80, n (%)	45/63 (71)	25/40 (63)	34/77 (44)	**.004**
Complication, n (%)				
Adverse events	1/69 (1)	5/45 (11)	12/81 (15)	**.017**
DVT occurrence	32/73 (44)	29/49 (59)	44/102 (43)	.15

[a]Bold indicated statistical significance ($P < .05$). ATRS, Achilles tendon Total Rupture Score; BMI, body mass index; DVT, deep venous thrombosis; NA, not applicable; TTS, time to surgery.

statistically significantly correlated with any of the outcome variables ($P > .05$).

Surgical Procedure

The surgical procedure was performed on an outpatient basis using uniform anesthetic and surgical techniques[22] as part of a standardized protocol for anesthesia and surgery, which was used in both RCTs[6,23] from which the patients in this study were selected.

With the patient in a prone position, local anesthetic was introduced to the skin, subcutis, and peritendinous space. Surgery was initiated with a medial incision over the Achilles tendon followed by a central incision through the fascia cruris and paratenon. Repair of the Achilles tendon was performed with a modified Kessler suture technique with two No. 1-0 polydioxanone threads, bringing the stumps of the ruptured Achilles tendon together end-to-end. The paratenon and fascia cruris were closed with a No. 3-0 Vicryl suture, and the skin was closed with a No. 3-0 Ethilon suture. To relieve potential postoperative pain, all patients were prescribed tablets containing paracetamol 500 mg and codeine 30 mg.

Postoperative Treatment

The postoperative treatment varied during the first 2 postoperative weeks due to the 2 RCTs incorporating partly different postoperative protocols. In the first RCT, patients were postoperatively randomized to either intermittent pneumatic compression under a brace (n = 69) or immobilization in a plaster cast (n = 70).[6] In the second RCT, patients were postoperatively randomized to either mobilization using an orthosis (n = 66) or immobilization in a plaster cast (n = 23).[23] Postoperative treatment was correlated with TTS ($R = 0.34$, $P \leq .001$); however, no statistically significant correlation was found between postoperative treatment and any of the outcome variables ($P > .2$). The lack of effect of the postoperative treatments used in both RCTs on patient outcome has also been demonstrated in 2 recently published studies.[1,23]

Assessment of Deep Venous Thrombosis

At 2 and 6 weeks postoperatively, all patients were screened for DVT in the operated leg by unilateral compression duplex ultrasonography. Two experienced ultrasonographers, blinded to the treatment regimens, performed all of the compression duplex ultrasonography procedures using a Philips CX 50 ultrasound machine (Philips Medical Systems). The standard procedure included evaluation of all deep proximal and distal veins, including muscle veins, as well as the great saphenous vein. The criteria for DVT diagnosis and the diagnostic procedure have been described earlier.[6,11] Proximal DVT was defined as a thrombosis that involved the popliteal vein or any more proximal veins, with or without involvement of the calf veins. Briefly, the DVT diagnosis was based on a transverse ultrasound compression test of the blood vessel and assessment of blood flow in the veins by color Doppler flow.

Assessment of Adverse Events

An adverse event was noted if the patient exhibited any of the following: perioperative complications, postoperative infections at 2 or 6 weeks of follow-up, or a rerupture at any time during the year following surgery. The rate of each of the complications described above was low, and they were therefore collapsed to 1 factor.

Patient-Reported Outcome Measures

Patients answered the validated ATRS questionnaire (Swedish, version 6) at the 12-month follow-up appointment to determine each patient's degree of symptoms. ATRS includes 10 specific questions (scored from 0 to 10, with 10 indicating no limitations). The maximal total score of the ATRS is 100, signifying the best possible outcome. A score of ATRS higher than 80 was considered an overall good subjective outcome.

Statistical Analysis

All data were entered in SPSS (IBM SPSS, version 24.0). The variables were summarized with standard descriptive

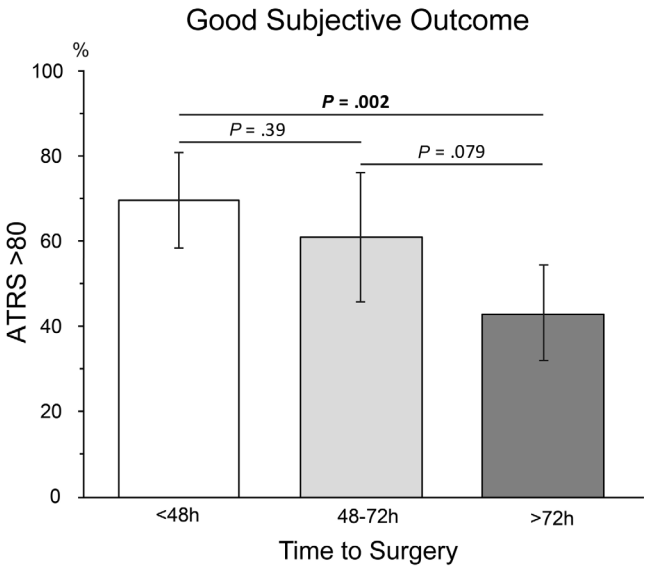

Figure 2. Good subjective outcome for each of the different time groups with 95% CI. ATRS, Achilles tendon Total Rupture Score.

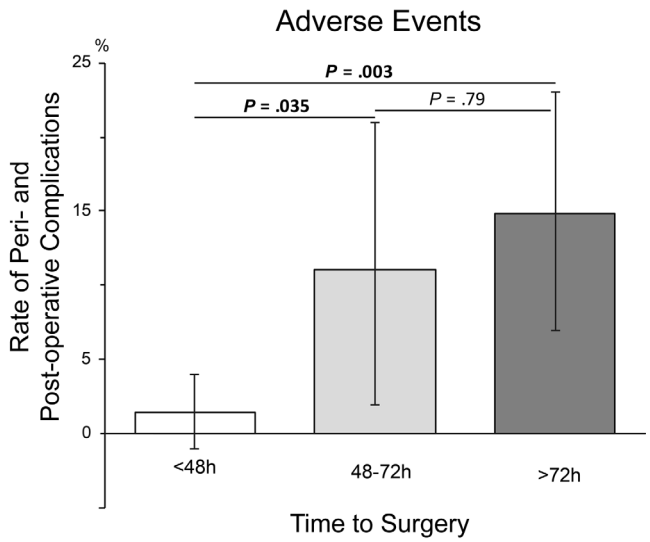

Figure 3. Adverse events for each of the different time groups with 95% CI.

statistics such as mean, SD, and frequency. Group differences were described with *P* values and 95% CI. All variables were checked for skewness. Correlations between different variables and outcome were expressed as Pearson correlation coefficients. Nonparametric Spearman rank correlation was used if a distribution was severely skewed. Comparisons between groups were performed with Pearson chi-square test for categorical variables and 1-way analysis of variance for continuous variables. The level of significance was $\leq$5% (2-tailed) for all analyses.

RESULTS

Patient-Reported Outcome

The prevalence of good subjective outcome (ATRS >80) at 1 year after ATR differed statistically significantly among the 3 TTS groups (*P* = .004; Table 1), with the best outcome in the group with short TTS. A good subjective outcome was observed in 71% (45/63) (95% CI, 60%-83%) of patients with short TTS (<48 hours), which was a significantly higher rate compared with patients with long TTS (>72 hours), where a good subjective outcome was seen in only 44% of patients (34/77) (95% CI, 33%-56%) (*P* = .002). In the patients with intermediate TTS (48-72 hours), a good subjective outcome was observed in 63% (25/40) (95% CI, 47%-78%) of the cases (Figure 2).

Adverse Events

The risk of exhibiting an adverse event differed significantly among the 3 TTS groups (*P* = .017; Table 1), with the highest prevalence in the long TTS group. Among patients with short TTS, 1.4% (1/69) exhibited an adverse event (95% CI, 1%-4%), which was significantly less compared with the intermediate TTS group (*P* = .035), where 11.1% (5/45) of patients exhibited an adverse event (95% CI, 2%-21%), and also significantly less than the group with long TTS (*P* = .003), where 14.8% (12/81) of patients exhibited an adverse event (95% CI, 7%-23%) (Figure 3).

Deep Venous Thrombosis

The risk of sustaining a DVT was not statistically significantly different among the 3 TTS groups (Table 1). In the short TTS group, the incidence of DVT was 43.8% (32/73) (95% CI, 32%-55%), which was lower compared with the intermediate TTS group, where 59.2% (29/49) (95% CI, 45%-73%) sustained a DVT. In the long TTS group, the occurrence of DVT was 43.1% (44/102) (95% CI, 33%-53%) (Figure 4).

DISCUSSION

The results of this study demonstrate for the first time that the timing of surgery in patients with acute ATR has a significant influence on patient-reported outcome at 1 year postoperatively. We show that reduced time to surgery improves patient-reported outcome and decreases the risk of adverse events after ATRs.

The most significant finding was to establish a 38% relative reduction (27% absolute reduction) in the number of patients achieving a good subjective outcome when surgery was delayed more than 72 hours compared with less than 48 hours. This finding reflects a substantial difference in patient-reported outcome, is corroborated[24] by conclusions from other surgical fields, and warrants changes in practice guidelines. Such a difference also suggests a new factor underlying the wide variation in ATR patient outcomes.

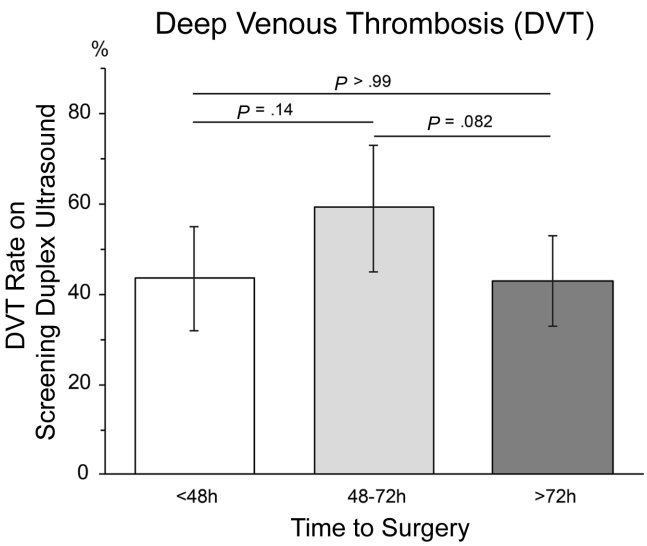

Figure 4. Deep venous thrombosis (DVT) incidence for each of the different time groups with 95% CI.

The observed discrepancy in good subjective outcome related to TTS may be partly explained when examining the questions in the validated ATRS questionnaire. An earlier study demonstrated that at 1 year after ATR surgery, 44% of patients experienced limitations due to pain in their limb and 48% of patients had limitations in walking on uneven surfaces.[22] It may be that prolonged TTS diminishes the healing potential of the patient, which may lead to an increased experience of pain. The established worse subjective outcome after more than 72 hours' delay may therefore be related to the biology of tendon healing.

After ATR injury, the first 72 hours are considered the induction phase of tendon healing.[2] The wound site is infiltrated with blood-derived cells acting as "traffic police," stopping bleeding, providing immune control, cleaning up tissue debris, and inducing further healing by the release of inflammatory mediators and growth factors. If a second trauma caused by surgery occurs after this induction phase is over, there may be a lack of healing inducing factors. Interestingly, human anterior cruciate ligament (ACL)–derived cells obtained during an acute rupture phase exhibit greater healing potential than the cells from a chronic rupture phase.[10] Another factor that has been discussed in regard to ACL surgery is that delayed TTS may cause retraction of the stumps.[13] Further studies, however, are needed to determine the exact mechanisms underlying the effects of TTS.

The second main finding of our study was the establishment of a positive relationship between longer TTS and increased number of adverse events. The observed time-dependent increase in, for example, postoperative infections may be related to a similar biological explanation to that which is described above. Hence, as the endogenous immune response declines within 72 hours after ATR injury, the risk of infection may increase with TTS.[2] Our findings are also in line with research from other surgical disciplines. For laparoscopic cholecystectomy, surgery within 2 days of presentation of acute cholecystitis has been demonstrated to yield the best outcome for patients.[24] In hip fracture surgery, it has been demonstrated that a delay of TTS increases the risk of both complications and mortality.[16,18,21,25]

The lack of a statistically significant difference in occurrence of DVT in combination with the nonlinear distribution of DVT among the 3 TTS groups suggests that other risk factors not fully accounted for in this study are of greater importance than TTS when predicting the prevalence of DVT.

Our findings of improved outcomes and reduced number of complications in patients undergoing surgery less than 72 hours after injury suggest that ATR surgery should not be delayed more than 3 days after injury. Since no current guidelines are available regarding TTS in patients with ATR, we propose a general recommendation to avoid system-related delays in ATR surgery and a specific recommendation to operate within 72 hours. Our recommendations are based on patients undergoing open ATR surgery but, according to principles of biological healing, also should be applicable to minimally invasive surgery. Our findings also suggest that all ongoing comparisons between nonoperative and operative treatments should take TTS into consideration and possibly only compare patients who undergo operation within 72 hours after injury.

Given our results that patients undergoing surgery less than 48 hours after injury exhibited the best results, it is essential that the perioperative care of ATR patients is adequate. The availability of operating rooms and support services needs to be optimized to avoid system-related delays in surgery. These findings also warrant future studies comparing patients treated surgically for ATR within 48 hours versus patients receiving nonoperative treatment.

Only 1 earlier publication, which included 65 patients, examined TTS for ATR.[17] That study mainly focused on functional outcome at 3 months and was unable to find any significant differences between patients grouped by TTS. Moreover, the earlier study by Park et al[17] did not include assessment of subjective outcome at 1 year after ATR surgery. Hence, until further data are available, we suggest the above-presented recommendations for surgically treated ATR patients. According to our knowledge, our study is by far the largest ATR cohort, comprising a total of 245 patients, investigating the relationship between TTS and patient outcome. Further research must confirm and extend our findings and understanding of TTS after ATR.

In a field where the amount of previous research is limited, our study adds highly relevant results to the current knowledge of how TTS may affect patient outcome. One potential limitation of our study might be that the patients in the different cohorts performed slightly different postoperative rehabilitation protocols. The rehabilitation protocols were, however, considered in the statistical analyses and did not affect the outcome. Another potential limitation would be that patients received different anesthetic and surgical procedures depending on TTS. However, since the same study protocol, regarding local anesthesia and surgical techniques, was used in both RCTs, we presume

all patients received similar treatment. The strengths of our study are the relatively large sample size and a meticulous assessment of the time from injury to surgery. One additional advantage of our study was that neither patients nor doctors had any influence on TTS, which was determined by availability of the operation theater.

The results of this study warrant prospective randomized trials to corroborate the current findings. However, we consider it unethical to evaluate the influence of a prolonged waiting time for surgery in ATR patients within the context of an RCT. Consequently, the best approach would be a cohort study including a large number of prospectively randomized patients, as was done in the present study.

CONCLUSION

Our study established that delayed TTS was associated with poorer patient-reported outcomes 1 year after surgery for an acute ATR. Prolonged TTS was further associated with a higher occurrence of adverse events. On the basis of these findings, we propose not to delay ATR surgery more than 72 hours.

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